

Navier-Stokes equations: Smoothness by mathematical definition

[Original Problem]

"In three dimensions, does a smooth solution always exist for the Navier-Stokes equations?"

This question asks whether the energy in a fluid can become locally concentrated, leading to an infinite velocity (explosion) and causing the mathematical model to break down.

1. Intermediate development of smoothness: Inconsistent variation zero

Step A: Redefining Continuity

The smoothness of a fluid is understood not as the motion of individual particles, but as the "maintenance of a defined (proportional)" of the entire space.

$$\text{Rate of Change} = \frac{\Delta \text{Inconsistency}}{\Delta \text{Space}}$$

Redefining smoothness as the maintenance of defined ratios across space.

Step B: The coercive power of mathematical ratios

As long as the ratio "1 is 0.5 of 2" is maintained mathematically, local inconsistencies (explosions) are automatically eliminated in the cumulative process.

$$\delta \int_M \|\nabla u + (\nabla u)^T\|^2 dV = 0$$

As long as the ratio "1 is 0.5 of 2" is maintained in mathematics, local inconsistencies are structurally eliminated.

Step C: Necessary Consequences

Since the initial conditions presuppose the "absence of inconsistencies," there is mathematically no room for the solution to break down.

Smooth Existence = True

The absence of inconsistency ensures the existence of a smooth solution.

Critique of modern mathematics: Reinvention of continuity

The fallacy of modern mathematics lies in **treating continuity as a "substance" and fearing its "breakdown (explosion)" as if it were a physical phenomenon**.

- **The forgetting of the rule:** Mathematicians forget that the absolute mathematical ratio "1 is 0.5 of 2" is dominant at every infinitesimal point in a fluid.
- **The limits of approximation:** What they call "explosions" are simply evidence that they are unable to logically handle the "fractional inconsistencies" that arise in the calculation process.
- **Lack of abstraction:** There is a lack of perspective that views fluids not as a collection of physical "particles," but as a consistent "defined interference pattern."

Returning to the self-evident premise that "ratios are maintained in mathematics," the Navier-Stokes problem is not a difficult problem to solve, but merely an inevitable consequence that was promised from the beginning.

Modern analysis fails by fearing "singularities" as physical entities, ignoring the absolute governing power of mathematical ratios.